Simultaneous Equations

1. Let s = the cost of one salad roll
   p = the cost of a pie

$$3s+4p = 20.56$$
$$s = 1.16+p$$

using classpad
$s = \$3.60$
$p = \$2.44$

A salad roll costs \$3.60 and a pie costs \$2.44

2. $$x+y = 12$$
$$3x-y = 16$$

$x=7$, $y=5$

3. Let x and y represents the two numbers.
$$x+y = 38$$
$$x-y = 24$$
the two numbers are 31 and 7

4. Let x and y represent two numbers of page for each book
$$x+y = 500$$
$$x = 3y-12$$

The number of pages for one of the books 128 while the number of pages for the other is 372.

5. Let l = length of the paddock
   w = width of the paddock
$$2l+2w = 140$$
$$l = w+10$$

The dimension of the paddock is 40 m by 30 m.

6. Let f = the number of 5¢ coins
   t = the number of 10¢ coins
$$f+t = 18$$
$$0.05f+0.1t = 1.65$$

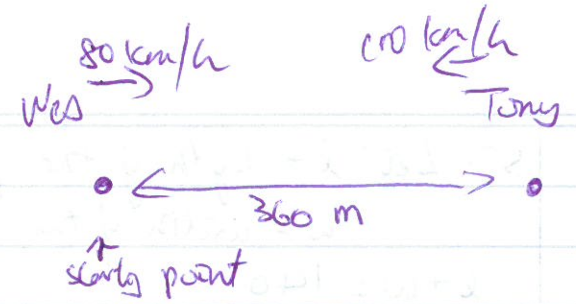
She received 3 5¢ coins and 15 10¢ coins.

7. Let
m = Michelle's present age
y = Yesmin's present age.

	Michelle age	Yesmin age
2 years ago	$m-2$	$y-2$
Now	M	y

$$2M-5 = y$$
$$2(y-2) = 10+3(M-2)$$
$$2y-4 = 10+3M-6$$
$$2y = 3M+8$$
$M=18$, $y=31$

Michelle's present age is 18
Yesmin's present age is 31



8.

distance = speed x time

Wes is moving away from his house and Tony is moving towards Wes' house.

Let d = distance in km from Wes' house
t = time taken in hours

a) Wes:
$d = 80t$

Tony:
$d = 300 - 100t$

b) solving using desmos.
They will meet after 2 hours.

9. Let b = speed of the boat
c = speed of the current

$3(b-c) = 27$ $\quad 3b - 3c = 9$
$2(b+c) = 30$ $\quad 2b + 2c = 30$

$b = 12$
$c = 3$

The speed of the boat is 12 km/h and the speed of the current is 3 km/h